

Novel Exchange Coupled Topological Interfaces for Magnetic Recording Applications

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Rich phenomena arise at novel magnetic interfaces, which have become a forefront for quantum materials. Breakthroughs have been fueled by improving capabilities of realizing and manipulating quantum behavior when emerging magnetic order, sizable spin-orbit coupling and topologically nontrivial states cooperate at the atomic level. Here we present recent progress in interfacial magnetism when topological insulator (Bi,Sb)₂Te₃ is coupled to ferromagnetic (Cr₂Te₃), antiferromagnetic (Mn(Bi,Sb)₂Te₄/MnTe) and altermagnetic (CrSb) layers. We highlight nontrivial spin textures and dynamics that are enabled by synergy of complex crystalline phases, competing orders, atomically engineered defects and symmetry breaking. We envision deeper understanding of coupled magnetism and topology to boost application of magnetic topological interfaces as memory and logic.

Index Terms—Spintronics, magnetic topological insulator, Weyl semimetal, proximity effect.

I. INTRODUCTION

Magnetic materials have continuously driven the evolution of information technologies through increasing control of interactions at reduced length scales. Demands for faster, denser, and more energy-efficient architectures are now pushing research toward atomically engineered thin films and quantum heterostructures. At the same time, condensed matter physics has entered a new regime where topology, spin-orbit coupling, symmetry breaking, and emergent magnetic order are intertwined. As shown in Fig. 1, beyond conventional ferro- and antiferromagnets, new altermagnets provide additional pathways for manipulating spin textures and anisotropic transport. Particularly exciting opportunities emerge when these magnetic phases are interfaced with topological materials, generating unique phenomena inaccessible in conventional magnetic systems. Advances in epitaxial synthesis, interface engineering, and nanoscale characterization now enable complex magnetic topological interfaces with atomic precision.

Here we highlight our recent progress on exchange-coupling topological insulator (TI) (Bi,Sb)₂Te₃ with ferro- (Cr₂Te₃ [1]), antiferro- (Mn(Bi,Sb)₂Te₄/MnTe [2]), and altermagnetic (CrSb [3]) layers. These systems provide a versatile platform for exploring emergent interfacial magnetism, symmetry-driven spin splitting, chiral spin textures, and coupled topological states relevant to future spintronic, magnonic, and magnetic recording technologies. We envision that deeper understanding of coupled magnetism and topology at engineered interfaces may ultimately provide new routes toward ultralow-power memory, logic, and nonvolatile information processing architectures [4].

II. METHODS

Our approach combines theory-guided materials design, atomic-scale synthesis, and multimodal characterization to investigate exchange-coupled magnetic topological interfaces relevant to next-generation spintronic technologies [4].

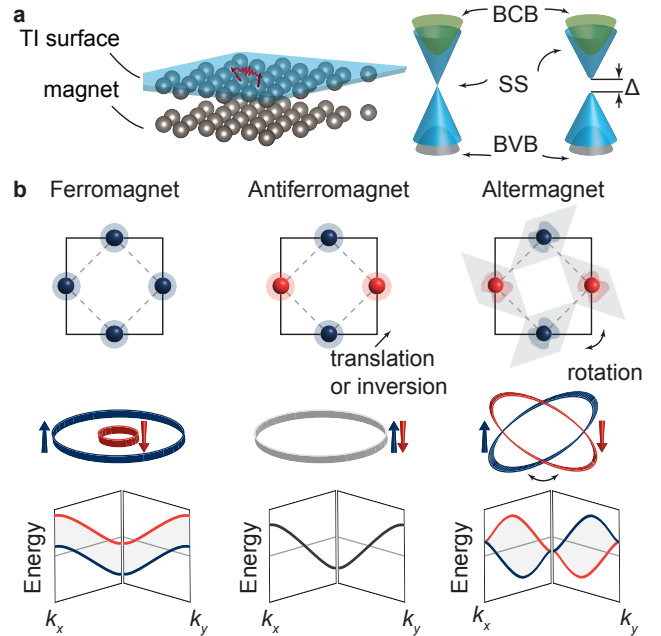


Fig. 1. Novel magnetic topological interfaces. **a** Proximity coupled magnetic topological insulator interfaces. An exchange gap develops in the Dirac-like surface states (SSs) in the gap of bulk conduction (BCB) and valence (BVB) bands, allowing chiral edge transport. **b** Typical magnetic orders ranging from ferromagnetic (left), antiferromagnetic (middle) to altermagnetic (right) [6].

First-principles density functional theory (DFT) calculations incorporating spin-orbit coupling and Berry curvature analysis were used to study electronic structure, magnetic interactions, and topological characteristics of candidate heterostructures. These calculations provide insight into symmetry breaking, interfacial coupling, and emergent spin textures that guide materials selection and interface engineering [5].

Magnetic topological thin films and heterostructures were grown using molecular beam epitaxy (MBE) carefully optimized to achieve atomically sharp interfaces [7,8]. *In situ* reflection high-energy electron diffraction (RHEED) was used to monitor epitaxial growth and surface reconstruction, while angle-resolved photoemission spectroscopy (ARPES) directly probed the evolution of band structures and surface states.

Structural characterization was carried out using X-ray diffraction (XRD) and scanning transmission electron microscopy (STEM). Buried interfaces and depth-sensitive magnetism were investigated through polarized neutron reflectometry (PNR) and grazing incidence inelastic neutron scattering (GIINS), enabling quantitative determination of layer thickness, interface roughness, chemical depth profiles, and interfacial magnetic configurations with nanometer resolution.

Nanodevice fabrication and comprehensive thermoelectric transport measurements were performed under variable temperature, magnetic field, dielectric gating, and angular configurations to study anomalous Hall response, spin-charge coupling, and emergent phenomena.

III. RESULTS

Coupling TI with ferro-, antiferro-, and altermagnetic layers enables strong proximity effects between ordered spins and topological electronic states. As shown in Figs. 2, PNR reveals that magnetization in quasi-2D ferromagnetic Cr_2Te_3 is significantly enhanced when interfaced with $(\text{Bi,Sb})_2\text{Te}_3$, demonstrating that topological surface states actively reinforce magnetic order through interfacial exchange coupling.

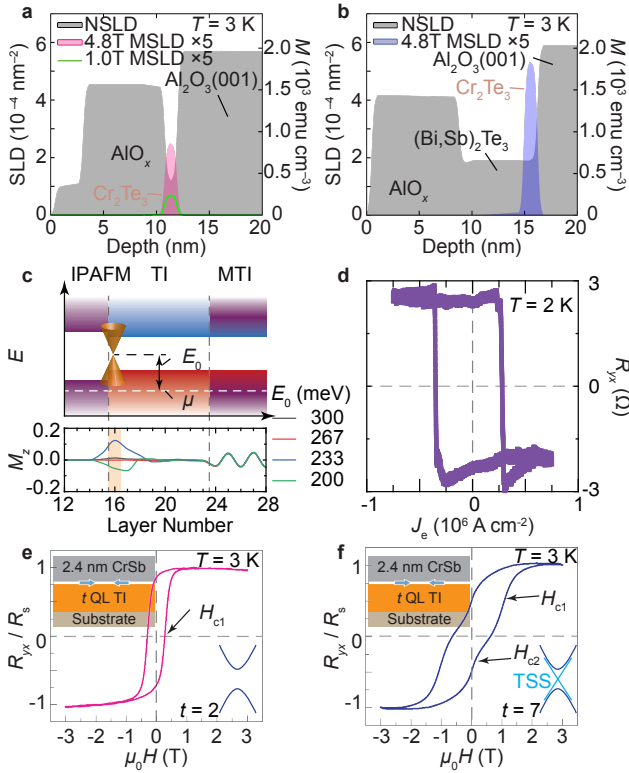


Fig. 2. Proximity magnetic topological interfaces. **a, b** Ferromagnet–topological insulator (TI). Polarized neutron reflectometry depth profile of magnetic (MSLD) and nuclear (NSLD) scattering length densities reveal TI surface induced enhancement of magnetism in Cr_2Te_3 (**a**) when coupled to TI (**b**). **c, d** Antiferromagnet–TI. **c** Quantum magnetic simulations suggest proximity coupled in-plane antiferromagnet (IPAFM)/TI/magnetic topological insulator (MTI) heterostructures (top) stabilize out-of-plane magnetization M_z at TI surfaces (bottom). **d** Electric current J_e induced spin-orbit torque type switching of Hall resistance R_{yx} in MnBi_2Te_4 -TI/ MnTe heterostructures. **e, f** Altermagnet–TI. Magnetic field dependent $R_{yx}(H)$ profile (normalized by saturation resistance R_s) reveals that CrSb can efficiently couple to topological surface state (TSS) when TI thickness exceeds critical transition thickness of 4 quintuple layers (QLs), opening a surface exchange gap as corroborated by the emergence of two-step hysteresis loop (from Ref. [6]).

More complex emergent behavior arises in antiferro- and altermagnetic topological heterostructures. In natural hybrids of $\text{Mn}(\text{Bi,Sb})_2\text{Te}_4$ -($\text{Bi,Sb})_2\text{Te}_3$ / MnTe , interfacial magnetism enables a magnetic field-free spin-orbit torque switching up to 200 K. Meanwhile, in altermagnet-TI $\text{CrSb}/(\text{Bi,Sb})_2\text{Te}_3$ bilayer lattice-mismatch-induced symmetry breaking enables pronounced anomalous Hall effects.

IV. CONCLUSION

Exchange coupled topological interfaces are emerging as a promising platform for next-generation magnetic recording technologies. Recent advances allow atomic-scale control of complex magnetic topological heterostructures. Looking forward, deterministic engineering of interfacial magnetism and topology may enable ultralow-power memory, spin logic, and other future information technologies while continuing to uncover new quantum phenomena.

ACKNOWLEDGMENT

H.C. acknowledges the support of the Army Research Office (ARO) under Grant Number W911NF-25-1-0215, Canada Research Chairs (CRC) Program, the Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery Grant RGPIN-2024-06497, ALLRP 608284-25, 613902-25, 613904-25, the University of Toronto's Acceleration Consortium (Canada First Research Excellence Fund, CFREF), Canada Foundation for Innovation (CFI), Ontario Research Fund (ORF), the Digital Research Alliance of Canada (alliancecan.ca) and Compute Ontario (computeontario.ca).

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